

Effects of Perceived Transport Integration: Evidence from Thameslink's Addition to the London Tube Map

We have substantial evidence that consumers have limited attention and use heuristics and imperfect signals when making decisions in many markets. We have much less evidence, however, on how relevant such behaviour is in high-stakes decisions such as buying a house. This dissertation shows that house prices in London are sensitive to a change in a highly imperfect signal of accessibility: the London Tube map. In January 2021, Thameslink services were added to the map despite no accompanying change in physical rail service. Using a difference-in-differences framework with monthly transaction data from HM Land Registry, I estimate that the Tube map addition is associated with a statistically significant 2.8 percent increase in house prices within 1 kilometre of treated Thameslink stations, relative to comparable properties in a 1.5–3 kilometre ring around treated stations. The results are robust to alternative control groups. Consistent with an informational mechanism, the effect is larger in outer fare zones (3.8 percent) than in inner fare zones (2.0 percent). These results suggest that limited attention to publicly available transport information can affect housing valuations, and that low-cost informational interventions can matter by reducing informational frictions.

I. INTRODUCTION

Housing markets are typically assumed to reflect underlying fundamentals such as transport accessibility, local amenities, and neighbourhood characteristics. In this view, buyers process available information and prices reflect the true value of these attributes. Yet a large body of behavioural research suggests that individuals do not fully process all available information and instead rely on salient signals and heuristics when making decisions (Simon 1955). Whether such behaviour persists in high-stakes settings such as housing markets remains an open question.

One implication of this is that buyers may not evaluate transport accessibility purely based on objective measures, but instead rely on simplified or incomplete representations of the transport network. This points to a distinction between physical accessibility and perceived accessibility (Lättman, Olsson, and Friman 2016). Physical accessibility refers to the actual transport service available, such as number of railway lines and their service frequency. Perceived accessibility refers to how connected a place appears to buyers when they decide whether to buy a house. The

* Generative AI (Claude and ChatGPT) were used in this project to help with coding and editing of writing. I verified the accuracy of the code and made sure that AI did not generate its own ideas and arguments.

two do not always coincide. An area can have good rail connections in practice but still look less accessible if those connections are not shown clearly on the maps buyers rely on.

This distinction between physical and perceived accessibility points to a gap between the transport economics and behavioural economics literatures. The transport economics literature usually assumes that house prices reflect nearby transport infrastructure (e.g. Gibbons and Machin 2005; Bowes and Ihlanfeldt 2001), and that buyers process the relevant features of the transport network. By contrast, the behavioural economics literature on bounded rationality (Simon 1955) and rational inattention (Sims 2003) suggests that people often rely on shortcuts when gathering and processing information. In London, one of the most important shortcuts for gathering public transport information is the Tube map.

This paper addresses this gap by exploiting the January 2021 addition of Thameslink, a National Rail service, to the Tube map. No new infrastructure was built and no service improvements occurred, yet this simple visual change may have altered how Londoners perceived the public transport connectivity of the areas served by Thameslink, especially in south London. If so, it suggests that homebuyers do not fully process all available transport information and instead place weight on a salient shortcut.

This is especially interesting because buying a home is one of the most important financial decisions households make, so one might expect buyers to collect and process transport information carefully. If a simple map inclusion still shifts house prices in that setting, the implication is important: limited attention is not just a feature of small, low-stakes decisions, but can also shape decisions in major housing transactions.

This research question matters for urban policy. If part of the property premium comes from perception rather than physical service, policymakers may have access to a relatively low-cost way to influence spatial patterns of demand: changing the information shown on transit maps, without the capital cost of building new rail lines. More broadly, the Thameslink case provides a way to test whether perceived transport integration has an independent effect on house prices, separate from the effect of actual transport improvements.

The empirical design compares properties within 1 kilometre of treated Thameslink stations with properties in a 1.5–3 kilometre ring around the same stations, before and after the January 2021 map change. The baseline estimates imply a 2.8 percent relative price increase for properties within 1 kilometre of treated stations. The effect is larger in outer fare zones than in inner fare zones. The estimated effect also remains consistent across a range of robustness checks, including specifications that use alternative fixed-effects structures and alternative control groups based on other National Rail operators.

Taken together, these findings suggest that housing markets respond not only to actual transport infrastructure, but also to how clearly that infrastructure is presented to buyers.

The remainder of the dissertation proceeds as follows. Section II reviews the related literature and explains the paper's contribution. Section III describes the Thameslink map change as a natural experiment. Section IV introduces the data, Section V sets out the empirical strategy, Section VI

presents the main results and robustness checks, Section VII discusses threats to identification and policy implications, and Section VIII concludes the analysis.

II. LITERATURE AND CONTRIBUTION

The starting point for this dissertation is the literature studying the effects of transport infrastructure investments, which shows that better accessibility is reflected in nearby property values. Gibbons and Machin (2005) find price gains of roughly 9–10 percent around the Jubilee Line Extension and the Docklands Light Railway stations, while Diao, Leonard, and Sing (2017) found an 8.6 percent increase in private housing values around Singapore's Circle Line. Similar evidence is found by Bowes and Ihlanfeldt (2001) for rail transit access in Atlanta. This literature establishes that transport matters for house prices, but it usually studies settings in which actual service changes or new services are introduced. As a result, it does not cleanly separate the value of physical accessibility from the value of being seen as connected to the wider transport network, or what the transport literature would describe as perceived accessibility (Lättman, Olsson, and Friman 2016).

This distinction is important in London, where the Tube map is more than a wayfinding tool. A small literature shows that it shapes both how Londoners understand the city and how they use the transport network. Vertesi (2008) shows that the map helps people locate places within London, and areas omitted from it might seem remote or less fully part of the city, even when they have usable rail services. Guo (2011) provides quantitative evidence that the map affects behaviour directly, showing that routes that appear longer on the map are chosen less often even after controlling for actual travel time. Taken together, these studies suggest that the Tube map influences perception and behaviour, but they do not show whether those effects are large enough to matter in housing market decisions.

One way to understand this is through the broader literature on bounded rationality and rational inattention. Simon (1955) argues that decision-makers rely on heuristics rather than fully processing all available information, while Sims (2003) and Matějka and McKay (2015) formalise the idea that attention is costly. In the housing market, this logic is especially plausible because buyers face a complex decision involving transport links, schools, neighbourhood quality, and future resale value. In principle, homebuyers could research timetables and compare all available transport options. In practice, they may rely on simpler public signals.

A growing housing literature applies this general idea to the housing market and shows that such signals can affect prices even when the underlying fundamentals do not change. Figlio and Lucas (2004) show that the housing market responds significantly to the information on government introduced school grade classifications, even when accounting for the underlying variables used to construct those classifications. Brounen and Kok (2011) find that consumers incorporate the energy-efficiency classifications into house prices. These studies suggest that housing markets respond not only to fundamentals, but also to how easy those fundamentals are to notice and interpret.

This dissertation brings these ideas together. It contributes to the literature studying the effects of transport infrastructure investments by isolating a change in perceived transport integration from any change in physical service. It contributes to the housing market literature by providing further evidence that a simple public signal can affect prices even in a high-stakes decision. More broadly, it provides evidence that limited attention matters in housing valuations, not just in smaller everyday decisions.

III. THE NATURAL EXPERIMENT

In December 2020, Transport for London (TfL) announced that Thameslink services would be added to the Tube map from the following month. The revised map then began appearing in January 2021 across several formats, including maps displayed in stations and on trains, the pocket Tube map, the TfL website, and the widely used TfL Go app. In total, 47 Thameslink stations were added to the Tube map in the 2020 announcement.¹ Crucially, this was a purely informational change: no new station opened, no new payment method was introduced, and no service expansion accompanied the map addition². The only relevant change is that these stations now appeared on the most widely recognised transit map in London.

London has seen other Tube map changes, including cases where former National Rail routes were brought into the London Overground network and added to the Tube map. Those settings are less suitable for identifying a pure informational effect because map inclusion was bundled with wider changes in branding, management, and operational integration, and in some cases with changes in ticketing or service quality. In those cases, any housing market response would be difficult to attribute to map visibility alone. The Thameslink case is more useful precisely because the map change occurred without a comparable change in physical service or fare integration.

Most stations added in this natural experiment were concentrated in south and south-east London. This was especially important for south London because that part of the city had long had much less Tube map coverage than other parts of London, despite extensive National Rail services. Figure 1 provides a simplified illustration of this change by comparing a pre-treatment Tube map with a later version showing Thameslink's inclusion.³ For a buyer relocating to London from abroad, places such as Deptford, Plumstead, or Catford may not have looked like plausible

¹This count refers only to stations that newly appeared on the Tube map as part of the Thameslink addition. Stations that already appeared on the map because they connected with existing Tube lines, such as London Bridge, are not treated here as newly added stations.

²The map change coincides with COVID-19, when rail services operated under reduced timetables. These reductions were widespread across operators rather than specific to Thameslink and later returned toward normal levels. To the extent that such changes were similar across treated and control areas, they are unlikely to bias the difference-in-differences estimates, although some residual variation cannot be ruled out.

³Figure 1 is a simplified illustration based on the 2019 and 2026 Tube maps. Because of TfL licensing constraints, it is not an exact reproduction of the official map and is intended only to show the broad change in Thameslink's visibility.

commuting options if they did not appear on the Tube map. The map change made those locations easier to recognise as part of London's rail network.

This natural experiment has two important identification advantages. First, there is little reason to expect anticipation. The map change was announced in December 2020 without prior notice and implemented the following month. Second, the rationale for treatment assignment helps rule out some confounding stories. In response to a Freedom of Information request, TfL explained that Thameslink was added because it functions as a cross-London service in a way comparable to other Tube lines, unlike operators such as South Western Railway whose routes mainly branch out from terminal stations. This is important because it implies that treatment status was tied to network structure rather than to a contemporaneous judgement that Thameslink areas were uniquely affected by the pandemic or by local regeneration. This natural experiment therefore offers a clear test of whether the salience of transport information matters for property prices.

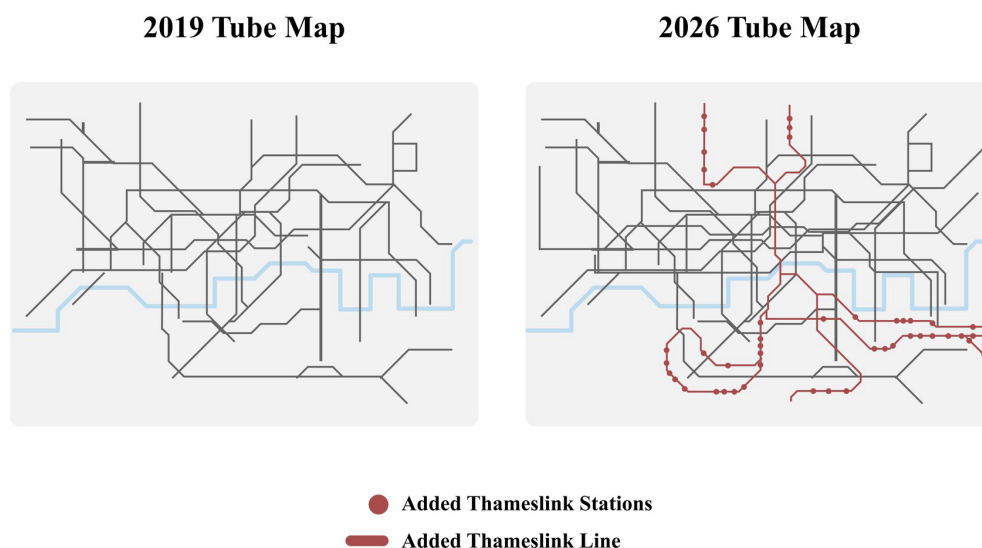


FIGURE 1. SIMPLIFIED ILLUSTRATION OF THE LONDON TUBE MAP BEFORE AND AFTER THAMESLINK'S INCLUSION.

IV. DATA

Studying the effects of the Thameslink treatment requires data at a very fine spatial scale. I use HM Land Registry Price Paid Data from 2016 to 2025. This dataset records all residential property transactions in England and Wales, providing transaction price, date, postcode, and address details. The dependent variable is the natural logarithm of the transaction price. Properties are identified

by the unique combination of postcode, Primary Addressable Object Name (PAON), and Secondary Addressable Object Name (SAON), and the sample is restricted to Greater London.

Postcodes are geocoded using the Ordnance Survey Code-Point dataset (British National Grid coordinates), enabling precise distance calculations between properties and stations. Property-level characteristics — property type, built form, total floor area, number of habitable rooms, construction age band, tenure, and extension count — are obtained from the Energy Performance Certificate (EPC) database and matched to transactions using postcode, PAON, and SAON.

Station locations are obtained from the Office of Rail and Road (ORR) station attributes dataset. Underground and DLR station coordinates are used to exclude properties within 1 kilometre of any existing Tube station.

I include local authority-level data on net additional dwellings from the ONS housebuilding statistics as a control variable and use it as a proxy for local development activity. Because the announcement was sudden and the treatment itself was relatively modest, it is unlikely that property developers could respond quickly enough for new construction to drive the short-run results, so this control is unlikely to absorb part of the treatment mechanism.

V. EMPIRICAL STRATEGY

A. Model Specification

Intuitively, adding Thameslink to the Tube map may make nearby areas appear better connected. This should increase demand for housing in those areas and therefore raise prices. To test this hypothesis, I use a difference-in-differences approach to examine whether houses near added stations experienced faster price growth than comparable houses further away. This approach is consistent with the literature studying the effects of transport infrastructure investments, which typically relies on before-and-after variation combined with fine local controls to separate the effect of transport changes from pre-existing differences across locations (e.g. Gibbons and Machin 2005; Gibbons et al. 2016; Ahlfeldt and Feddersen 2018; Hodgson 2018). The baseline estimating equation is:

$$p_{it} = \beta_1(Treat_i \times Post_t) + X_{it}'\beta_2 + \gamma_c + \delta_{s \times y} + \mu_m + \varepsilon_{it},$$

where p_{it} is the natural logarithm of the transaction price of property i at time t ; $Treat_i$ equals 1 if the property is within 1 kilometre of a treated Thameslink station and 0 if it lies in the baseline control group, defined as properties 1.5–3 kilometres from the same treated station; $Post_t$ equals 1 for transactions from January 2021 onwards and 0 for transactions up to December 2020; X_{it} includes the time-varying additional-dwellings control at the local authority level and a set of property characteristics from the EPC data (property type, built form, total floor area, number of habitable rooms, construction age band, tenure, and extension count); γ_c denotes postcode fixed effects; $\delta_{s \times y}$ denotes station-by-year fixed effects; μ_m denotes calendar month fixed effects; and ε_{it} is the error term.

The coefficient of interest is β_1 , which captures the differential change in log house prices for treated properties relative to control properties after Thameslink's addition. Given the log-linear specification, β_1 can be interpreted as the approximate percentage change in house prices as a result of the treatment. Under the hypothesis of the paper, this coefficient is expected to be positive.

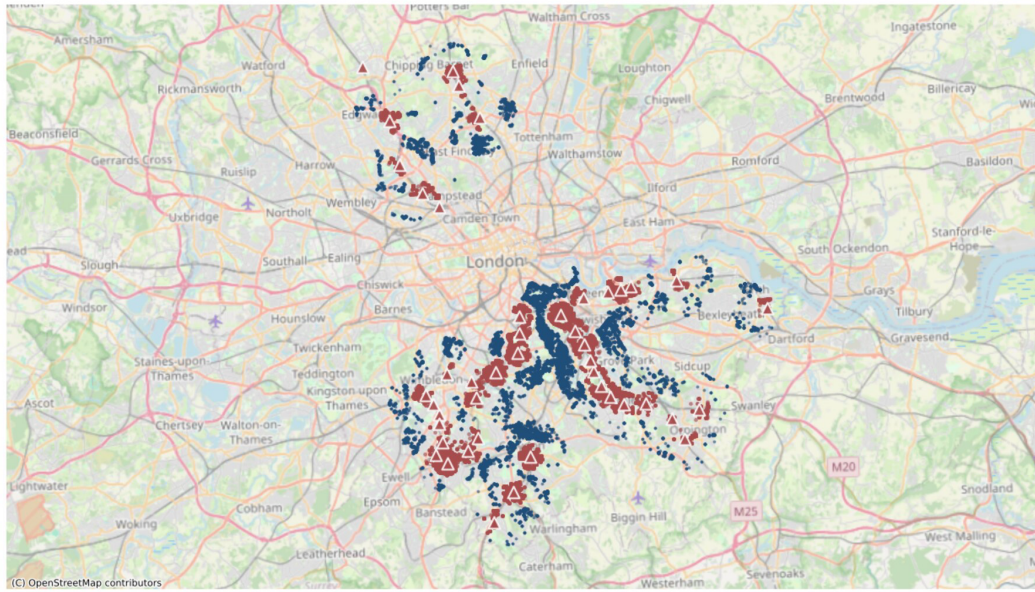
Standard errors are clustered at the postcode level to account for spatial correlation in house prices within postcodes and serial correlation in prices over time for properties in the same postcode.

B. Treatment and Control Groups

The December 2020 announcement added 47 Thameslink stations to the map. Two of these, Dartford and Swanley, are outside Greater London and are excluded from the estimation sample.⁴ The final regression sample therefore retains 45 treated Thameslink stations. The full station list is provided in Appendix A. The treatment group thus consists of properties within 1 kilometre of these 45 treated stations. The 1 kilometre threshold is intended to capture properties within a plausible walking distance of a station and is broadly consistent with the transport economics literature (e.g. Gibbons and Machin 2005).

The baseline regression adopts a ring control-group method. The ring control group consists of properties located 1.5–3 kilometres from treated Thameslink stations. These properties are in the same general area as treated properties and therefore share similar local economic conditions, demographic characteristics, and housing market trends, but they are too far from the treated stations to plausibly benefit from the map addition. This logic is consistent with studies that compare units close to treatment with units slightly further away in the same local area. In transport, Diao, Leonard, and Sing (2017) use treated and control zones around Singapore MRT stations in a spatial difference-in-differences design. Outside transport, Linden and Rockoff (2008) compare houses very close to sex offenders with slightly further houses in the same neighbourhood. A 500-metre buffer zone between the 1 kilometre treatment boundary and the 1.5 kilometre control boundary is included to reduce potential spillover effects from the treatment. Figure 2 shows the spatial structure of the baseline design, including the treated Thameslink stations, treated houses, and the 1.5–3 kilometre ring control.

⁴Brent Cross West is not included in the treated set because it opened only in 2023, after the January 2021 treatment date, even though it appears on the Tube map at the time of writing.



- ▲ Added Thameslink Stations
- Control Houses (1.5-3km of Treated Stations)
- Treated Houses (<1km of Treated Stations)

FIGURE 2. BASELINE TREATMENT AND RING-CONTROL AREAS AROUND TREATED THAMESLINK STATIONS

To ensure that the analysis focuses on houses for which the Thameslink treatment was likely to matter, I exclude all properties within 1 kilometre of any London Underground or DLR station from both the treatment and control groups. The reason is that these properties were already effectively represented on the Tube map before Thameslink was added, so the informational effect of Thameslink's inclusion is likely to be limited. This restriction therefore focuses the analysis on properties for which the map change should plausibly matter.

The descriptive comparison in Table 1 supports the use of this ring control. In the restricted estimation sample, the treatment and ring-control groups have very similar pre-treatment and post-treatment mean log prices, and they also share similar characteristics such as flat share, floor area, rooms, and extension count. Overall, the descriptive evidence supports the idea that the ring control is a locally comparable counterfactual rather than a fundamentally different part of the London housing market.

TABLE 1—DESCRIPTIVE COMPARISON OF TREATMENT AND RING-CONTROL GROUPS IN THE RESTRICTED ESTIMATION SAMPLE

	<i>Treatment (< 1 km)</i>	<i>Ring control (1.5–3 km)</i>
Log price (pre)	12.66 (0.49)	12.68 (0.57)
Log price (post)	12.71 (0.43)	12.72 (0.53)
Observations	21,053	21,812
Postcodes	2,282	2,476
Flat (percent)	90.9	89.7
Floor area (sq. m.)	64.5	65.6
Habitable rooms	2.86	2.91
Owner-occupied (percent)	56.6	58.5
Private rented (percent)	33.7	33.4
Social rented (percent)	8.8	7.1
Pre-1930 stock (percent)	28.1	25.4
Extension count	0.13	0.13

Notes: Standard deviations are reported in parentheses. The restricted sample consists of postcodes with at least one pre-treatment and one post-treatment transaction. Properties within 1 kilometre of any existing London Underground or DLR station are excluded.

C. Fixed Effects Structure

The baseline specification includes postcode fixed effects, station-by-year fixed effects, and month fixed effects. The choice of postcode fixed effects rather than individual property fixed effects represents a deliberate trade-off between controlling for unobserved heterogeneity and maintaining adequate sample size. With property fixed effects, my sample would be limited to houses that transact at least once in both the pre-treatment and post-treatment periods. Given that the average holding period for residential property in London is approximately 8–10 years, the probability of observing the same property transacting in both a five-year pre-period and a five-year post-period is relatively low. This constraint would sharply reduce the sample size. This is also in line with Gibbons and Machin (2005), who use very fine spatial fixed effects in a London housing setting rather than relying solely on repeat sales at the individual property level.

Postcode fixed effects relax this requirement by only requiring a postcode to contain transactions in both periods, while different individual properties within that postcode can contribute to the pre and post observations. This is useful because postcodes are still very small geographic units, but they typically contain about 10–20 dwellings (Royal Mail Postcode Address

File (PAF)). The cost is that within-postcode variation in property characteristics could bias the estimate if the composition of transacting properties changes differentially between treated and control postcodes after the treatment. For example, if the map addition increased the share of transactions involving higher-quality properties within treated postcodes, the average transaction price could rise even without any change in underlying property prices. To address this concern, the regression includes detailed EPC property characteristics as controls, capturing the most important dimensions of within-postcode property heterogeneity including property type, built form, floor area, number of rooms, construction age, tenure type, and the number of extensions. These controls are among the strongest determinants of property prices within a given location and should absorb much of the composition effect on observables.

Restricting the sample to postcodes with at least one pre-treatment and one post-treatment transaction reduces the number of observations only modestly. Table 2 shows that the filter reduces the sample size without changing average prices or house characteristics, which suggests that the identifying sample is not being created by selecting unusually expensive or unusually cheap areas. For ease of presentation, the same treatment and control definitions set out in Section V.B are applied in this table.

Station-by-year fixed effects ($\delta_{s \times y}$) are possible because all properties in the baseline specification are defined relative to treated Thameslink stations. Each property is assigned to its nearest treated station, and identification comes from comparing properties close to that station with properties slightly further away in the same station area and year. These fixed effects therefore absorb shocks common to the broader station catchment in a given year, such as local regeneration, neighbourhood demand shifts, or station-specific service changes, without absorbing the treatment effect itself. Calendar month fixed effects (μ_m) capture seasonal patterns in the housing market.

TABLE 2—DESCRIPTIVE COMPARISON BEFORE AND AFTER THE POSTCODE PRE/POST FILTER

	Treatment		Control	
	<i>Before filter</i>	<i>After filter</i>	<i>Before filter</i>	<i>After filter</i>
Log price	12.68 (0.47)	12.68 (0.46)	12.70 (0.57)	12.70 (0.55)
Observations	22,355	21,053	23,486	21,812
Postcodes	2,704	2,282	2,967	2,476
Flat (percent)	90.1	90.9	88.7	89.7
Floor area (sq. m.)	64.8	64.5	66.3	65.6
Habitable rooms	2.89	2.86	2.94	2.91
Owner-occupied (percent)	56.1	56.6	57.6	58.5
Private rented (percent)	34.0	33.7	34.1	33.4
Social rented (percent)	9.0	8.8	7.3	7.1
Pre-1930 stock (percent)	28.4	28.1	25.9	25.4
Extension count	0.14	0.13	0.14	0.13

Notes: Standard deviations of log price are reported in parentheses. The sample covers properties within 1 kilometre of treated Thameslink stations (Treatment) and within 1.5–3 kilometres of treated Thameslink stations (Control). The "Before filter" columns use the full sample of transactions; the "After filter" columns restrict to postcodes with at least one pre-treatment and one post-treatment transaction.

D. Identification Assumptions

The validity of the DiD design rests on two key assumptions.

Parallel Trends. The fundamental assumption is that, absent the treatment, house prices near treated Thameslink stations would have followed the same trajectory as house prices in the control group. This assumption is inherently untestable, but the event study analysis provides suggestive evidence by examining whether pre-treatment trends are parallel.

No Major Developments near Treated Stations. The design assumes that no major regeneration or development projects were implemented near treated stations at the same time as the map change. Station-by-year fixed effects help by absorbing shocks common to the broader station area in a given year, such as area-wide demand shifts, changes in local economic conditions, or improvements that equally affect both the near-station and outer-ring areas around the same station. However, they would not fully remove bias from very local shocks concentrated mainly in the treated station area, such as a regeneration project, major new amenity, or redevelopment located very close to the station but not affecting the outer ring. I have conducted background

research on the affected areas and found no evidence of major concurrent developments of this type.

E. Robustness

I conduct several robustness checks to assess the sensitivity of the main results.

Alternative Fixed Effects Structure. As a robustness check on the spatial level at which fixed effects are defined, I estimate alternative specifications using output area fixed effects and station-ring fixed effects. Output areas sit above postcodes in the census geography and contain small clusters of neighbouring postcodes. Station-ring fixed effects group observations into 500-metre distance bands around each treated station. Together, these checks test whether the main result is robust to using alternative local fixed-effects structures rather than relying only on postcode fixed effects.

Alternative Treatment Radii. I also vary the treatment radius while keeping the same 1.5–3 kilometre ring control group. This checks whether the main result depends on the exact distance threshold used to define treatment.

COVID Exclusion Check. Because the January 2021 treatment date overlaps with the pandemic period, I also estimate a specification that excludes 2021 and 2022 and redefines the post period to begin in 2023.

Placebo Treatment Date. As a placebo test, I assign a placebo treatment date of January 2018, restricting the sample to the pre-treatment period only (2016–December 2020), to check whether the DiD design spuriously detects an effect in the absence of a real treatment.

VI. EMPIRICAL RESULTS

A. Main Results

Table 3 presents the results for the baseline specification. The baseline ring specification (0–1 kilometre treatment versus 1.5–3 kilometre control) yields a coefficient of 0.0281, implying that treated properties experienced approximately 2.8 percent higher price growth relative to the control group after the Tube map change.

TABLE 3—BASELINE DIFFERENCE-IN-DIFFERENCES RESULTS

<i>Variable</i>	<i>(1) Baseline ring DiD</i>
Treated $\times$ Post	0.0281*** (0.0100)
Observations	42,865
Postcode fixed effects	Yes
Station $\times$ year fixed effects	Yes
Month fixed effects	Yes
EPC controls	Yes
Additional dwellings control	Yes

Notes: The dependent variable is the natural logarithm of the transaction price. The coefficient on Treated $\times$ Post is the difference-in-differences estimate of the effect of the January 2021 Tube map change on house prices within 1 kilometre of treated Thameslink stations, relative to the 1.5–3 kilometre ring control. Standard errors clustered at the postcode level are reported in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

The 2.8 percent magnitude of this estimate is also economically plausible. The effect is noticeably smaller than the roughly 9–10 percent price gains found by Gibbons and Machin (2005) for houses near newly built transport infrastructure in London, which is what one would expect given that the treatment here is only a map change rather than a physical improvement in service. Figlio and Lucas (2004) found that Florida's introduction of A–F school grades generates a 3–4 percent price premium for houses near schools just above a grade threshold, relative to those just below, despite near-identical underlying school quality. Like the Tube map addition, the school grade is a government-provided label that changes how buyers evaluate a neighbourhood characteristic they could have researched independently. The similarity in magnitude across these settings suggests that a 2–4 percent effect is plausible for an informational treatment in the housing market.

The event-study graph in [Figure B3](#) in Appendix B does not indicate a clear violation of the parallel-trends assumption. The pre-treatment coefficients remain fairly close to zero overall and do not show a systematic divergence before 2021, which is consistent with parallel trends. Post-treatment coefficients are generally positive, but the estimates are imprecise and most confidence intervals are wide. This is unsurprising because the event-study specification splits the sample across multiple coefficients, which reduces power and makes the estimates noisier than the main DiD. Taken together, the baseline regression and the event-study pattern are generally consistent with a positive treatment effect.

B. Robustness Checks

I next test whether the result is robust to alternative specification choices. Table 4 summarises the robustness checks.

TABLE 4—ROBUSTNESS CHECKS

<i>Variable</i>	(1) <i>SWR</i>	(2) <i>Southeastern</i>	(3) <i>Output Area FE</i>	(4) <i>Station × Ring FE</i>	(5) <i>< 500 m</i>	(6) <i>< 1.5 km</i>	(7) <i>Post- 2023</i>	(8) <i>Placebo (Jan 2018)</i>
Treated × Post	0.0327*** (0.0124)	0.0327* (0.0181)	0.0274** (0.0117)	0.0279** (0.0123)	0.0232 (0.0143)	0.0174** (0.0088)	0.0284** (0.0126)	0.0005 (0.0157)
Observations	35,039	23,820	46,161	47,235	29,050	53,476	32,670	21,426
Postcode FE	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Station × year FE	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EPC controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional dwellings control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the natural logarithm of the transaction price. Columns (1) and (2) replace the ring control with operator-specific controls (South Western Railway and Southeastern). Columns (3) and (4) replace postcode fixed effects with output area and station-ring fixed effects. Columns (5) and (6) vary the treatment radius while keeping the 1.5–3 kilometre ring control. Column (7) excludes 2021 and 2022 and defines the post period as 2023 onwards. Column (8) uses a placebo treatment date of January 2018 with the sample restricted to 2016–December 2020, before the Tube map change. Standard errors clustered at the postcode level (or output area level in column (3)) are reported in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

1. Using Another Operator as Control Group

A first concern is that the positive baseline regression coefficient may still capture factors unrelated to the Tube map change itself. For example, if Londoners had generally become more willing to pay to live near rail stations after 2021, the baseline coefficient could be positive even without any special effect from Thameslink's inclusion on the Tube map. Another concern is that the ring control may be partly exposed to spillovers if the 1.5–3 kilometre control range also benefited from the map change. To address these concerns, I replace the local ring control with properties near stations on alternative National Rail lines that are not included on the Tube map. I chose South Western Railway (SWR) and Southeastern Railway because they are located in broadly comparable parts of London south of the River Thames, do not overlap too heavily with Thameslink routes, and have many stations within Greater London. This is also why Southern Railway is not used as a control group: its route overlaps too much with Thameslink, which would

blur the treatment and control distinction. I also exclude houses within 3 kilometres of treated Thameslink stations from the SWR and Southeastern Railway control groups. The spatial distribution of these alternative control groups is shown in [Figures B1](#) and [B2](#) in Appendix B. According to columns (1) and (2) of Table 4, both coefficients remain positive and are of very similar magnitude to the baseline estimate, if anything marginally larger, with the SWR estimate significant at the 1 percent level and the Southeastern estimate significant at the 10 percent level because of its smaller sample size.

Nevertheless, the ring specification remains the preferred design for two main reasons. First, the ring control is less exposed to unrelated local developments. The treatment here is an informational shock with a relatively small expected effect, so house price changes caused by other local developments could easily overshadow the effect of the map change. Using operators in different parts of London as controls exposes the estimates to exactly this risk: SWR and Southeastern stations are not located near the treated Thameslink stations, face different local shocks, and do not allow the same station-by-year comparison as the ring specification. A regeneration project or new retail development in an area served heavily by SWR, for example, could shift prices near many SWR control stations for reasons unrelated to Thameslink's inclusion on the map. Second, the ring control group is more comparable to the treatment group in observable characteristics than the SWR or Southeastern control groups, as shown in Appendix [Table B1](#). SWR and Southeastern control groups differ substantially from the treatment group in terms of average price levels, tenure composition, and pre-1930 housing share. This reinforces the idea that the ring control provides the tighter local counterfactual. For that reason, the operator-control regressions provide supportive evidence, but they do not replace the baseline local design.

2. *Output Area Fixed Effects*

A second concern is that postcode fixed effects may create coverage issues in more suburban areas. Postcodes are very small units, often containing around 15 dwellings (Royal Mail Postcode Address File (PAF)), so the pre/post filter can disproportionately exclude thinly traded locations. In addition, some suburban postcodes contain very few properties, which makes them less likely to survive the pre/post filter and enter the sample. I address this by replacing postcode fixed effects with output area fixed effects. Output areas are the lowest-level census geography in England and Wales and consist of small clusters of neighbouring postcodes, so they relax this constraint while still controlling for very local neighbourhood conditions (Office for National Statistics, Statistical Geographies). In this specification, I also cluster standard errors at the output area level. Table 5 clearly shows the sample-size differences: the estimation sample rises from 42,865 in the postcode fixed-effects regression to 46,161 with output area fixed effects. As shown in column (3) of Table 4, the coefficient remains very close to the baseline estimate. The trade-off is that output areas are less fine than postcodes, so they absorb less unobserved heterogeneity in house characteristics.

3. *Station Ring Fixed Effects*

A related way to address the coverage issue of postcode fixed effects is to use station-ring fixed effects. This approach divides the area around each treated station into distance bands and requires

pre-treatment and post-treatment observations within each station-ring band rather than within each postcode. For example, properties 0–500 metres from a given treated station share one fixed effect, properties 500–1000 metres from the same station share another, and so on. Table 5 shows that this further increases the estimation sample to 47,235. As shown in column (4) of Table 4, the coefficient remains close to the baseline estimate. Taken together, the results are very robust to choosing postcode, output-area, or station-ring fixed effects.

TABLE 5—SAMPLE SIZES UNDER DIFFERENT FIXED-EFFECTS STRUCTURES

<i>Fixed Effect Level</i>	<i>Sample Size</i>
Full sample (no pre/post filter)	47,276
Postcode pre/post sample	42,865
Output area pre/post sample	46,161
Station $\times$ ring pre/post sample	47,235

Notes: Sample sizes are reported under different fixed-effects structures. The full sample excludes the pre/post filter; the other rows apply a pre/post filter at the corresponding geographic unit.

4. *Different Treatment Radius*

A third concern is whether the result is sensitive to the exact 1 kilometre treatment boundary. In a dense transport network such as London's, one might argue that 1 kilometre is too wide and captures properties only weakly exposed to the salience effect; alternatively, the effect might extend beyond that threshold. Columns (5) and (6) of Table 4 address this by narrowing the treatment radius to 500 metres and widening it to 1.5 kilometres, while keeping the same 1.5–3 kilometre ring control. Both coefficients remain positive, although only the wider treatment radius is statistically significant at the 5 percent level.

The pattern is not perfectly monotonic, since one might expect the 500 metre coefficient to exceed the 1 kilometre baseline. However, the 500 metre regression also uses a smaller sample and has a larger standard error. The more expected pattern is that the 1.5 kilometre treatment radius produces a smaller coefficient than the 1 kilometre baseline, which is consistent with the distance-decay pattern often found in the transport literature, including the Jubilee line evidence from Pogonyi, Graham, and Carbo (2021). Overall, these checks suggest that the result is not driven by the exact 1 kilometre cutoff.

5. *Excluding COVID Period*

A fourth concern is that the January 2021 timing overlaps with the pandemic period. I address this by excluding 2021 and 2022 and redefining the post period to begin in 2023. As shown in column (7) of Table 4, the coefficient remains positive and statistically significant at the 5 percent level. This suggests that the COVID period transaction data is not affecting the main results.

6. Placebo Treatment Date

A final concern is that the DiD design may be picking up a pre-existing trend in property prices rather than a genuine treatment effect. As a placebo test, I assign a placebo treatment date of January 2018 and restrict the sample to the pre-treatment period only (2016–December 2020), so that the placebo pre-period covers 2016–2017 and the placebo post-period covers 2018–2020. If the design were generating false positives, we would expect to find a significant coefficient even when no real treatment occurred. As shown in column (8) of Table 4, the estimated coefficient is statistically indistinguishable from zero. This null result suggests that the design is not mechanically generating positive treatment effects.

C. Heterogeneity and Mechanisms

This section provides some evidence on the mechanism through which house prices were affected by Thameslink's inclusion on the Tube map.

TABLE 6—HETEROGENEITY AND MECHANISM RESULTS

<i>Variable</i>	<i>(1) Inner zones</i>	<i>(2) Outer zones</i>	<i>(3) Transaction volume</i>
Treated $\times$ Post	0.0199 (0.0136)	0.0380** (0.0148)	−0.0326 (0.0245)
Outcome	Log price	Log price	Log monthly transactions
Observations	42,865	42,865	10,234

Notes: Columns (1) and (2) split the treatment effect between inner fare zones (Zones 2–3) and outer fare zones (Zones 4–6). Column (3) uses the natural logarithm of monthly transaction counts as the dependent variable, with station-by-treatment group, year, and calendar month fixed effects. Standard errors clustered at the postcode level (columns 1–2) and at the station-by-treatment group level (column 3) are reported in parentheses. *** Significant at the 1 percent level. ** Significant at the 5 percent level. * Significant at the 10 percent level.

1. Heterogeneity by Fare Zones

If the mechanism is informational, the effect should be larger where Thameslink stations were less visible or less salient before the map change. The fare-zone heterogeneity specification addresses this by splitting the treatment effect between inner-zone stations (Zones 2–3) and outer-zone stations (Zones 4–6). As shown in columns (1) and (2) of Table 6, the estimated coefficient is 0.0199 for inner-zone stations and 0.0380 for outer-zone stations, implying effects of around 2.0 percent in inner London and 3.8 percent in outer London.

This pattern is consistent with the informational interpretation of the results. The Tube map had much larger gaps in the outer areas, especially in South London. By contrast, inner-zone Thameslink stations were already surrounded by many other stations that appeared on the Tube

map, so the addition of Thameslink there provided much less new information. The larger coefficient in outer zones therefore suggests that the treatment mattered most where the informational gap on the map was biggest.

2. *Transaction Volume*

The final question is whether the map addition affected market turnover as well as prices. We might expect to see an increase in transaction volume as more buyers entered the market for these areas. The transaction volume regression has a similar structure as the baseline ring regression, using the log of monthly transactions as the dependent variable, station-by-treatment group fixed effects, year fixed effects, and calendar month controls.

As shown in column (3) of Table 6, the estimated coefficient is -0.0326 , which is statistically indistinguishable from zero. This suggests no detectable effect on transaction turnover. This is reassuring from an identification perspective, because it suggests that the result is unlikely to be driven by a sudden increase in owners listing homes near treated stations, which could otherwise have changed the composition of transactions.

VII. DISCUSSION

A. *Other Threats to Identification*

Several threats to identification remain. First, the Stable Unit Treatment Value Assumption (SUTVA) may be violated if the treatment status of one location affects outcomes in other locations, including the control group. If housing demand is reallocated from control areas to treated areas, the coefficient could be biased upwards. A range of papers in the transport literature documents this type of effect: Redding and Turner (2015) emphasise that transport changes can reallocate activity across space, while Pogonyi, Graham, and Carbo (2021) show this mechanism for the Jubilee Line Extension, where areas close to stations gained activity while nearby untreated areas lost activity. In the present setting, however, this concern is likely to be much weaker than in studies of major transport infrastructure, since housing search is not typically confined to a narrow geographic band. Because the control group is at a very local level, it is less plausible that households who would otherwise have chosen to live 1.5-3 km from an added station instead saw the updated Tube map and relocated to the within-1 km area. A more plausible scenario is that a buyer who would otherwise have chosen a home near a different Tube station, or in another well-known part of London, instead chose a home near a newly mapped Thameslink station. That type of displacement would not bias the baseline estimate through the ring control. For this reason, any SUTVA violation is likely to have only a limited effect on the estimated treatment coefficient.

Second, some unobserved housing characteristics are not recorded in the EPC data. Even after controlling for floor area, rooms, age, tenure, built form, and extensions, there may still be quality differences such as interior refurbishment, furnishing, or finish that change over time. If treated areas began transacting better-furnished properties after the map change, measured prices could rise even without a true treatment effect. This concern therefore cannot be ruled out completely,

but it is reduced by the detailed EPC controls, the postcode fixed effects, and the absence of any detectable increase in transaction volume near treated stations.

Third, the Price Paid Data itself has some limitations. It excludes transactions that are not standard market sales, including gifts, some transfers linked to divorce, compulsory purchases, right-to-buy sales at a discount, and other special cases not recorded as ordinary value transactions. This means the dataset does not cover every property transfer in London. Houses never transacted during the sample period also will not show up in the data. However, there is little reason to think that Thameslink's appearance on the Tube map would differentially affect the incidence of these excluded categories near treated stations relative to nearby controls. For that reason, this limitation is more likely to affect the overall representativeness of the sample than to bias the treatment effect.

B. Policy Implications

These results matter for urban and transport policy. The Thameslink case suggests that public policy may matter not only through physical changes such as new infrastructure, faster journeys or more frequent services, but also through the way transport information is presented. In this setting, a map change that did not involve new infrastructure or major service improvements was nevertheless associated with higher house prices near treated stations. This is consistent with the view that some locations may be disadvantaged not only by weaker transport links, but also by the fact that existing links are less visible or less easily understood by households.

At the same time, this result should not be interpreted as meaning that the Thameslink map change has made London better off as a whole. A more likely interpretation is that it shifts housing demand from one part of the city to another. A household moving to London might previously have chosen somewhere in east London or another familiar area, but instead choose a home near a newly mapped Thameslink station. In that sense, the main effect of the map change may be redistribution across places rather than a net gain for the city as a whole. Even so, this can still be useful from a policy perspective. If an area has already improved, for example through a major regeneration project or wider neighbourhood investment, making its transport links more visible may help direct attention toward it. In that case, the role of information policy is not to create value from nothing, but to shift demand toward places that policymakers want to promote.

VIII. CONCLUSION

This dissertation exploits the January 2021 addition of Thameslink to the London Tube map to test whether limited attention affects housing market valuations. Using a difference-in-differences design with postcode fixed effects and a local ring control, I find a statistically significant 2.8 percent price increase near treated stations in the baseline specification, with positive estimates persisting across a wide set of robustness checks.

The central contribution of the paper is to show that housing prices can respond not only to actual transport accessibility, but also to how easily that accessibility can be recognised. No new infrastructure was built and no major service improvement occurred, yet prices still rose when

Thameslink became more visible on the map that many buyers use to understand London transport. This suggests that at least some buyers do not fully process all available information on actual rail connectivity and instead place weight on a salient public shortcut.

The broader pattern of results is consistent with this interpretation. The effect is larger in outer fare zones, where Thameslink stations were less likely to be salient before the map change, while transaction volume does not respond detectably, suggesting that the main effect operates through valuations rather than market turnover. Taken together, these findings suggest that informational frictions can shape how buyers assess locations and therefore how house prices are capitalised.

More broadly, the paper suggests that the design and presentation of public transport information can have real economic effects. A map change is far cheaper than building new infrastructure, but in this case it still appears to have altered valuations in parts of London that had long been less visible on the Tube map. The wider implication is that transport policy may work not only through tracks, trains, and journey times, but also through attention, salience, and the way connectivity is presented to the public.

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APPENDIX A: TREATED THAMESLINK STATIONS IN THE FINAL REGRESSION SAMPLE

Elstree and Borehamwood	Mill Hill Broadway	Hendon
Cricklewood	West Hampstead Thameslink	New Barnet
Oakleigh Park	New Southgate	Loughborough Junction
Herne Hill	Tulse Hill	Streatham
Tooting	Mitcham Eastfields	Haydons Road
Wimbledon Chase	South Merton	Morden South
St Helier	Sutton Common	West Sutton
Sutton	Carshalton	Hackbridge
Deptford	Maze Hill	Westcombe Park
Charlton	Plumstead	Slade Green
Nunhead	Crofton Park	Catford
Bellingham	Beckenham Hill	Ravensbourne
Shortlands	Bromley South	Bickley
Petts Wood	Orpington	St Mary Cray
Coulsdon South	Purley	South Croydon

APPENDIX B: ADDITIONAL TABLES AND FIGURES

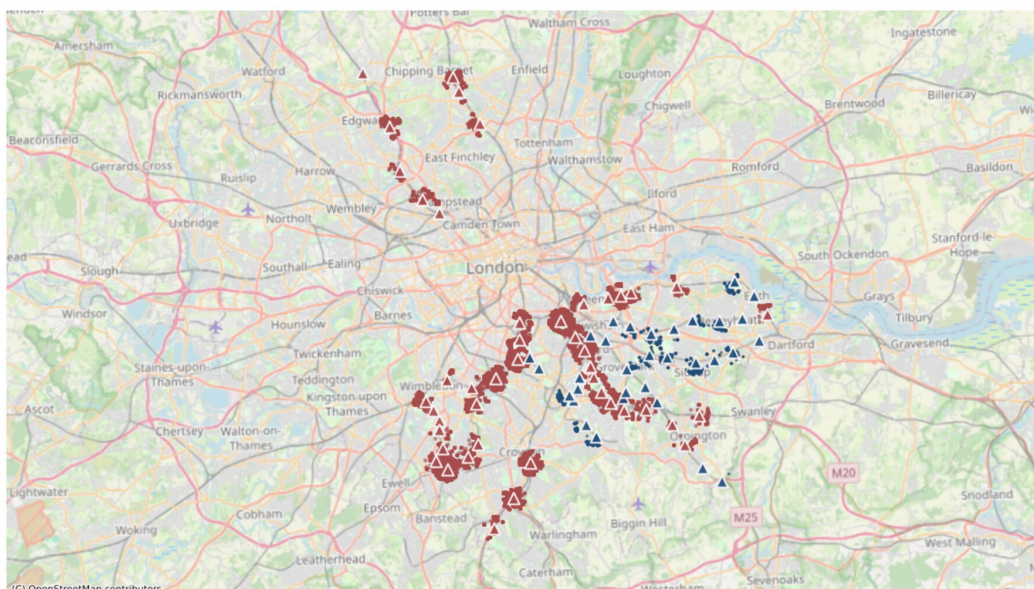
TABLE B1—BASELINE CHARACTERISTICS OF THE TREATMENT GROUP, RING CONTROL, AND OPERATOR CONTROL GROUPS IN THE RESTRICTED SAMPLE

	(1) <i>Treatment</i> (<i>< 1 km</i>)	(2) <i>Ring control</i> (<i>1.5–3 km</i>)	(3) <i>SWR control</i>	(4) <i>Southeastern control</i>
Log price (pre)	12.66 (0.49)	12.68 (0.57)	12.91 (0.64)	12.45 (0.49)
Log price (post)	12.71 (0.43)	12.72 (0.53)	12.89 (0.52)	12.44 (0.49)
Observations	21,053	21,812	13,975	2,750
Postcodes	2,282	2,476	1,477	268
Flat (percent)	90.9	89.7	93.3	88.1
Floor area (sq. m.)	64.5	65.6	66.9	61.9
Habitable rooms	2.86	2.91	2.78	2.82
Owner-occupied (percent)	56.6	58.5	59.6	74.5
Private rented (percent)	33.7	33.4	33.2	19.7
Social rented (percent)	8.8	7.1	6.2	4.1
Pre-1930 stock (percent)	28.1	25.4	18.6	9.2
Extension count	0.13	0.13	0.10	0.08

Notes: Standard deviations of log price are reported in parentheses. The SWR and Southeastern control groups use properties within 1 kilometre of stations on those National Rail operators that are not on the Tube map, excluding properties within 3 kilometres of any treated Thameslink station.



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- ▲ Added Thameslink Stations
- ▲ Control Southeastern Stations (Not Added)
- Treated Houses (<1km of Treated Stations)
- Control Houses (<1km of Control Stations)

FIGURE B2. THAMESLINK AND SOUTHEASTERN AREAS USED IN THE OPERATOR-CONTROL ROBUSTNESS CHECK

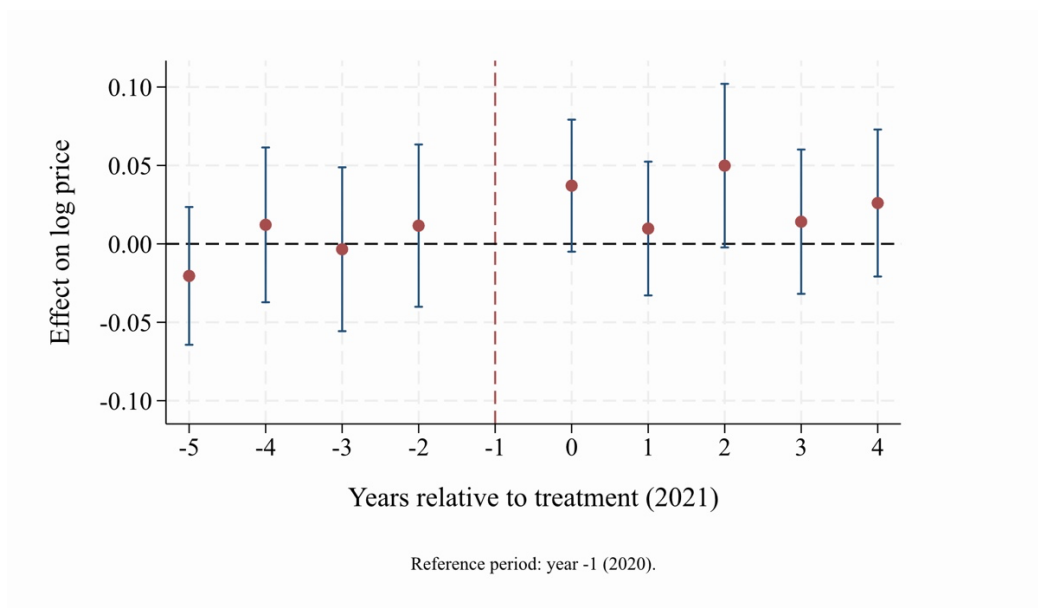


FIGURE B3. EVENT-STUDY ESTIMATES FOR THE BASELINE SPECIFICATION